

## MARKOVO, Russian Federation

WMO: 255510

Lat: 64.6742N

Lon: 170.4278E

Elev: 86

StdP: 14.65

Time Zone: 12.00 (E12)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -51.6      | -47.8 | -56.0                         | 0.2 | -51.5 | -52.3 | 0.3 | -47.7 | 18.9                  | 12.9 | 15.9 | 11.6 | 0.6                   | 230  | 0.765 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 19.3                   | 78.2            | 61.6 | 74.5 | 60.1 | 70.7 | 58.7 | 64.1                | 74.9 | 61.8 | 71.2 | 59.7 | 68.2 | 6.3                  | 230  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 59.7                            | 76.8 | 67.9 | 57.4 | 70.6 | 65.8 | 55.5 | 65.7 | 63.5 | 29.4          | 75.0 | 27.7 | 71.2 | 26.3 | 68.1 | 73.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |      |                    | n-Year Return Period Values of Extreme Temperature |           |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|-------|------|--------------------|----------------------------------------------------|-----------|------|------------|------|------------|------|------------|------|
| 1%                |      | 2.5% | 5%                         | Mean  |      | Standard Deviation |                                                    | n=5 years |      | n=10 years |      | n=20 years |      | n=50 years |      |
| Min               | Max  | Min  | Max                        | Min   | Max  | Min                | Max                                                | Min       | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  |                            | (e)   | (f)  | (g)                | (h)                                                | (i)       | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 15.7              | 13.2 | 11.4 | DB                         | -53.2 | 83.8 | 4.7                | 3.5                                                | -56.6     | 86.4 | -59.3      | 88.4 | -62.0      | 90.4 | -65.3      | 92.9 |
|                   |      |      | WB                         | -53.4 | 67.3 | 5.0                | 2.8                                                | -56.9     | 69.3 | -59.9      | 70.9 | -62.6      | 72.5 | -66.3      | 74.5 |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan    | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |       |      |
|------|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|      |                                                                                  | (d)         | (e)    | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |       |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 18.8   | -16.3 | -13.3 | -4.8  | 8.5   | 32.8  | 52.9  | 58.4  | 52.7  | 41.0  | 20.5  | 1.8   | -11.4 | (6)  |
| (7)  |                                                                                  | DBStd       | 30.01  | 18.37 | 17.67 | 14.82 | 12.86 | 9.34  | 7.07  | 6.18  | 6.01  | 7.36  | 11.76 | 16.53 | 20.14 | (7)  |
| (8)  |                                                                                  | HDD50       | 11944  | 2055  | 1771  | 1699  | 1246  | 533   | 50    | 7     | 41    | 281   | 913   | 1446  | 1902  | (8)  |
| (9)  |                                                                                  | HDD65       | 16892  | 2520  | 2191  | 2164  | 1696  | 997   | 366   | 216   | 381   | 720   | 1378  | 1896  | 2367  | (9)  |
| (10) |                                                                                  | CDD50       | 543    | 0     | 0     | 0     | 0     | 1     | 136   | 269   | 126   | 11    | 0     | 0     | 0     | (10) |
| (11) |                                                                                  | CDD65       | 16     | 0     | 0     | 0     | 0     | 0     | 2     | 13    | 1     | 0     | 0     | 0     | 0     | (11) |
| (12) |                                                                                  | CDH74       | 351    | 0     | 0     | 0     | 0     | 0     | 95    | 234   | 22    | 0     | 0     | 0     | 0     | (12) |
| (13) |                                                                                  | CDH80       | 47     | 0     | 0     | 0     | 0     | 0     | 9     | 37    | 1     | 0     | 0     | 0     | 0     | (13) |
| (14) | Wind                                                                             | WSAvg       | 4.7    | 4.0   | 4.6   | 4.9   | 4.9   | 5.0   | 5.4   | 4.7   | 4.4   | 4.6   | 4.7   | 4.8   | 4.7   | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 15.7   | 0.9   | 0.9   | 0.9   | 0.7   | 0.8   | 1.1   | 1.6   | 2.3   | 1.7   | 1.5   | 1.8   | 1.4   | (15) |
| (16) |                                                                                  | PrecMax     | 20.6   | 2.0   | 2.5   | 3.0   | 1.5   | 1.8   | 2.5   | 4.1   | 4.1   | 4.3   | 5.5   | 3.0   | 3.3   | (16) |
| (17) |                                                                                  | PrecMin     | 11.5   | 0.0   | 0.1   | 0.0   | 0.0   | 0.2   | 0.2   | 0.2   | 0.9   | 0.1   | 0.2   | 0.6   | 0.0   | (17) |
| (18) |                                                                                  | PrecStd     | 3.1    | 0.6   | 0.7   | 0.6   | 0.4   | 0.4   | 0.6   | 1.0   | 0.9   | 1.1   | 1.0   | 0.7   | 0.8   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 32.3  | 30.0  | 36.3  | 38.2  | 62.0  | 81.2  | 84.6  | 76.9  | 67.9  | 46.8  | 35.9  | 34.6  | (19) |
| (20) |                                                                                  | 0.4%        | MCWB   | 30.8  | 28.9  | 33.4  | 33.2  | 48.5  | 59.4  | 65.3  | 62.6  | 56.3  | 43.4  | 33.8  | 32.7  | (20) |
| (21) |                                                                                  | 2%          | DB     | 24.5  | 25.2  | 30.0  | 34.4  | 55.6  | 77.1  | 79.8  | 72.1  | 61.8  | 42.2  | 33.0  | 28.9  | (21) |
| (22) |                                                                                  | 2%          | MCWB   | 23.4  | 24.6  | 28.3  | 31.2  | 45.0  | 58.3  | 63.6  | 59.4  | 51.5  | 39.7  | 32.2  | 27.8  | (22) |
| (23) |                                                                                  | 5%          | DB     | 17.7  | 20.0  | 24.4  | 31.8  | 51.2  | 73.3  | 76.3  | 68.2  | 57.2  | 38.6  | 29.9  | 23.3  | (23) |
| (24) |                                                                                  | 5%          | MCWB   | 16.9  | 19.2  | 23.3  | 29.7  | 43.0  | 57.0  | 62.1  | 58.1  | 49.9  | 36.4  | 29.3  | 22.6  | (24) |
| (25) |                                                                                  | 10%         | DB     | 10.8  | 14.2  | 17.3  | 28.1  | 46.3  | 68.9  | 72.6  | 64.5  | 53.3  | 34.8  | 24.9  | 17.3  | (25) |
| (26) |                                                                                  | 10%         | MCWB   | 10.1  | 13.4  | 16.3  | 25.8  | 40.1  | 55.4  | 60.9  | 56.7  | 48.1  | 32.5  | 24.0  | 16.5  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 30.5  | 29.2  | 33.8  | 33.9  | 49.3  | 63.2  | 67.5  | 65.1  | 57.2  | 44.3  | 34.1  | 32.6  | (27) |
| (28) |                                                                                  | 0.4%        | MCD B  | 32.3  | 30.1  | 36.2  | 36.7  | 60.5  | 77.1  | 78.3  | 73.9  | 65.1  | 46.4  | 35.3  | 34.3  | (28) |
| (29) |                                                                                  | 2%          | WB     | 23.5  | 24.5  | 28.3  | 31.7  | 45.8  | 59.6  | 65.4  | 61.4  | 53.1  | 40.3  | 32.1  | 28.1  | (29) |
| (30) |                                                                                  | 2%          | MCD B  | 24.5  | 25.2  | 29.9  | 34.0  | 55.0  | 72.7  | 76.6  | 68.6  | 58.8  | 42.1  | 32.8  | 29.1  | (30) |
| (31) |                                                                                  | 5%          | WB     | 16.9  | 19.3  | 23.0  | 29.7  | 43.2  | 57.6  | 63.5  | 59.1  | 50.9  | 36.7  | 29.2  | 22.5  | (31) |
| (32) |                                                                                  | 5%          | MCD B  | 17.6  | 20.0  | 24.5  | 31.9  | 50.5  | 70.9  | 74.1  | 66.3  | 55.4  | 38.2  | 30.0  | 23.3  | (32) |
| (33) |                                                                                  | 10%         | WB     | 10.0  | 13.4  | 16.4  | 25.9  | 40.4  | 55.9  | 61.6  | 57.2  | 48.8  | 33.0  | 24.2  | 16.5  | (33) |
| (34) |                                                                                  | 10%         | MCD B  | 10.7  | 14.2  | 17.3  | 28.0  | 46.0  | 68.1  | 70.9  | 63.4  | 52.8  | 34.4  | 24.8  | 17.2  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | MDBR        | 15.8   | 18.1  | 20.1  | 20.1  | 14.8  | 19.3  | 19.3  | 18.1  | 16.6  | 13.7  | 14.7  | 14.9  | (35)  |      |
| (36) |                                                                                  | 5% DB       | MCD BR | 16.5  | 14.8  | 16.5  | 14.5  | 22.5  | 30.0  | 28.9  | 26.3  | 26.2  | 12.1  | 11.7  | 15.1  | (36) |
| (37) |                                                                                  | 5% DB       | MCWBR  | 16.3  | 14.6  | 15.0  | 11.7  | 13.2  | 13.7  | 15.0  | 15.1  | 17.1  | 9.3   | 11.4  | 14.7  | (37) |
| (38) |                                                                                  | 5% WB       | MCD BR | 16.7  | 14.5  | 16.3  | 13.4  | 20.9  | 26.0  | 25.1  | 22.0  | 20.5  | 9.7   | 11.7  | 15.0  | (38) |
| (39) |                                                                                  | 5% WB       | MCWBR  | 16.5  | 14.3  | 14.9  | 10.9  | 12.6  | 12.8  | 13.9  | 13.4  | 13.9  | 8.0   | 11.5  | 14.6  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.166  | 0.240 | 0.303 | 0.358 | 0.410 | 0.394 | 0.502 | 0.486 | 0.339 | 0.277 | 0.189 | 0.175 | (40)  |      |
| (41) |                                                                                  | taud        | 1.938  | 2.145 | 2.064 | 1.951 | 1.998 | 2.298 | 1.953 | 1.946 | 2.398 | 2.253 | 1.929 | 1.925 | (41)  |      |
| (42) |                                                                                  | Ebn at Noon | 129    | 214   | 240   | 251   | 249   | 260   | 220   | 208   | 234   | 194   | 103   | 40    | (42)  |      |
| (43) |                                                                                  | Edh at Noon | 8      | 18    | 31    | 45    | 47    | 36    | 50    | 45    | 23    | 16    | 7     | 3     | (43)  |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 45     | 271   | 830   | 1542  | 1790  | 1749  | 1568  | 1171  | 701   | 296   | 76    | 15    | (44)  |      |
| (45) |                                                                                  | RadStd      | 5      | 30    | 72    | 64    | 159   | 118   | 160   | 120   | 74    | 36    | 9     | 1     | (45)  |      |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (d)  | (e)                    | (f)   | (g)     | (h)    | (i)     | (j)   | (k)   | (l)         | (m)   |       |       |
| (46) | Station Only           | +2.05 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (0 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page